ChE 451 (Renewable Energy Technologies)
Department of Chemical Engineering
Fall 2019

Instructor:
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Office Hours: By appointment

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Prerequisite: course suitable for senior and graduate engineering, science, and business students.

Course outline
Part I: Sources of Renewable Energy
1) Solar
2) Wind
3) Biomass
4) Other sources

Part II: Energy Storage and Conversion
5) Batteries and Fuel Cells

Part III: The Hydrogen Economy
6) Hydrogen Production (reforming and electrolysis)
7) Hydrogen Storage and Transportation

Part IV: Case Studies (Applications)
8) Hybrid Wind/Solar Hydrogen Systems
9) Hybrid Desalination Systems: fuel cell/desalination, solar desalination

Text Books:
"Hybrid Hydrogen Systems for Stationary and Transportation Applications", Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3

Grade:
30% Weekly assignments (3-4 mini-projects: written essays and engineering design calculations)
30% Midterm Home Exam
40% Final Project: written report (20%) and oral presentation (20%)
Project Grading Criteria: originality, approach, content and relation to course material

Class Schedule

Week	Topic
1	Introduction (Renewable Energy and Sustainability)
2	Solar Energy Calculations and Solar Photovoltaics (PV) (*Discussion of Final Project*)
3	Building Integrated Photovoltaics
4	Wind Energy Basics and Applications *(Final Project Abstract Due)*
5	Bioenergy
6	Hydrogen Economy and Hydrogen Storage
7	Fuel Cells
8	Energy Storage for Smart Grid Applications
9	Geothermal Energy
10	Midterm *(Final Project Pitch)*
11	Hydrogen Economy: Production and Infrastructure
12	Optimization of Renewable Hydrogen Energy Systems
13	Control of Hybrid Fuel Cell/Battery Systems
14	Solar Desalination and Hybrid Fuel Cell/Desalination
15	Final (Oral Presentations)
16	Finals Week (written report due)